EXPLORATORY MEETING · ELLEN WAN & ANDREW HOFFMAN

Clay, microbes and metal

How clay hydration and iron chemistry gate microbially influenced corrosion, and whether a small BSL-1 pilot is a shared question



Hunter Kinder · independent researcher · October 2026

START HERE

Three questions decide whether this is a fit for us

1

For Ellen

Is a clay–EPS–metal interface study a fit for your biofilm and EPS work, and with which BSL-1 strains?

2

For Andrew

Which corrosion-product signatures (FeS, Fe_3O_4, Cu_2S) would you trust from Raman, SEM/EDS and XRD after 2–4 weeks?

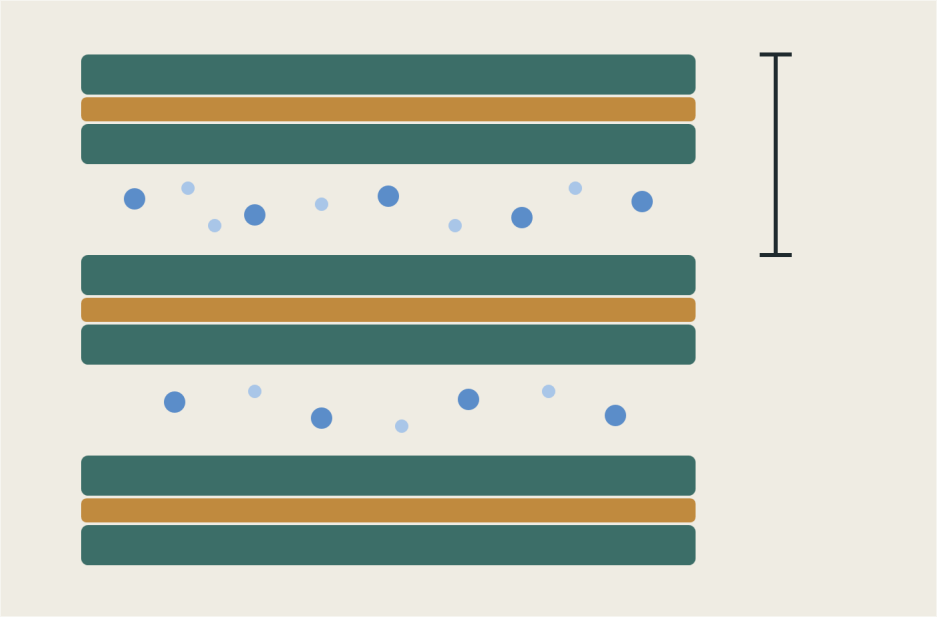
3

For both

What is the smallest result that would justify a 3–6 month study, and what would make you say no?

LITERATURE · MEASURED

Montmorillonite takes up water in discrete steps, and the cation sets which step



State	d001 spacing
Dry (0W)	≈10.0 Å
One water layer (1W)	≈12.6 Å
Two layers (2W)	≈15.2–15.3 Å
Three layers (3W, Na)	≈18–19.5 Å

At 40% RH, Na-montmorillonite sits near 12.5 Å (1W) and Ca-montmorillonite near 15.2 Å (2W). Only Na clays go on to osmotic swelling.

Ferrage et al. 2005a, Am. Mineral. 90:1358-1374; 2005b, Clays Clay Miner. 53:348-360. Schematic redrawn for this deck (green: silicate sheets; ochre: octahedral sheet; blue: cations and water). Bracket marks d001.

LITERATURE · MEASURED

Compaction shuts off sulfide production but leaves sulfate reducers alive

1740–1880

kg/m³ **wet** density where sulfide production collapses (≈1.2–1.4 dry)

10^5–10^6

SRB per litre still alive at 2000 kg/m³, where activity is near zero

0.954

water activity measured in situ at 1.50 g/cm³ **dry** (0.983 at 1.25)

Open question: 10–15 MPa of pressure alone did not suppress activity; water activity, pore size and heat are the competing explanations.

SKB TR-16-09 (Bengtsson et al.); Engel et al. 2023, mSphere; Bartak et al. 2024, World J Microbiol Biotechnol. SKB reports wet density; Grimsel reports dry.

LITERATURE · FIELD AND MODEL

Interfaces, not the bulk buffer, are where microbes grow and metal corrodes

Setting	What was seen	Evidence
Grimsel MaCoTe, 5 years	Outer bentonite up to 66% Desulfosporosinus; inner core unchanged	Field, measured
Mont Terri steel	1.5–2.25 µm/yr at 20 months, 0.75 µm/yr at 5.5 years; FeS linked to SRB	Field, measured
Copper canister	≤1.3 mm in 1 Myr; iron minerals trap ~93% of sulfide	Model
Copper without goethite	Corrosion ≈20× deeper	Model

Engel et al. 2023, mSphere; Necib et al. 2017, Swiss J Geosci; Burzan et al. 2022, Front Microbiol; Ma et al. 2022, Appl Geochem.

LITERATURE · LAB AND FIELD

Iron moves both ways: steel stains the clay, and microbes reduce the clay’s iron

Steel → clay

- Äspö ABM1: ≈100 µm corrosion layer, ≈10 mm iron-enriched clay, no new clay phases
- Grimsel FEBEX, 18 years: iron halos beyond 140 mm
- Swelling loss in the field: inferred, not measured

Microbes → clay

- Smectite → illite in 14 days at room temperature (abiotic: 300–350 °C)
- Fe(III) reduction raised CEC 0.82 → 1.5 meq/g in iron-rich clay, only 0.85 → 0.99 in montmorillonite
- Swelling pressure fell in every iron-bearing smectite tested

Hypothesis, not shown: microbial iron reduction loosens the barrier and speeds corrosion at the interface.

Wersin et al. 2015; Hadi et al. 2019; Kim et al. 2004, Science; Stucki & Kostka 2006; Gates et al. 1998.

EVIDENCE IS THIN HERE

Biofilm polymers enter clay interlayers and Ca^{2+} bridges EPS, but clay-specific data are thin

Evidence type	Finding
Direct (hectorite)	With P. syringae, d001 went 1.50 → 1.26 nm in 72 h, then 1.84 nm in about a week
Analogue (no clay)	64 mM Ca^{2+}: biofilm biomass 25–51× control, thickness 2.1 → 19.6 µm
Review	Clay–microbe interaction knowledge is “still very incomplete”
Hypothesis	Ca^{2+} released at the interface stabilises a clay–EPS–metal biofilm

This gap is where Ellen’s EPS expertise could matter most.

Alimova et al. 2009, Clays Clay Miner.; Xiong et al. 2022, Front Environ Sci; Cuadros 2017, Clay Miner. (CC BY 4.0).

THE QUESTION I WOULD MOST LIKE YOUR VIEW ON

Screening, chelation and bridging are three different things, and only one changes connectivity

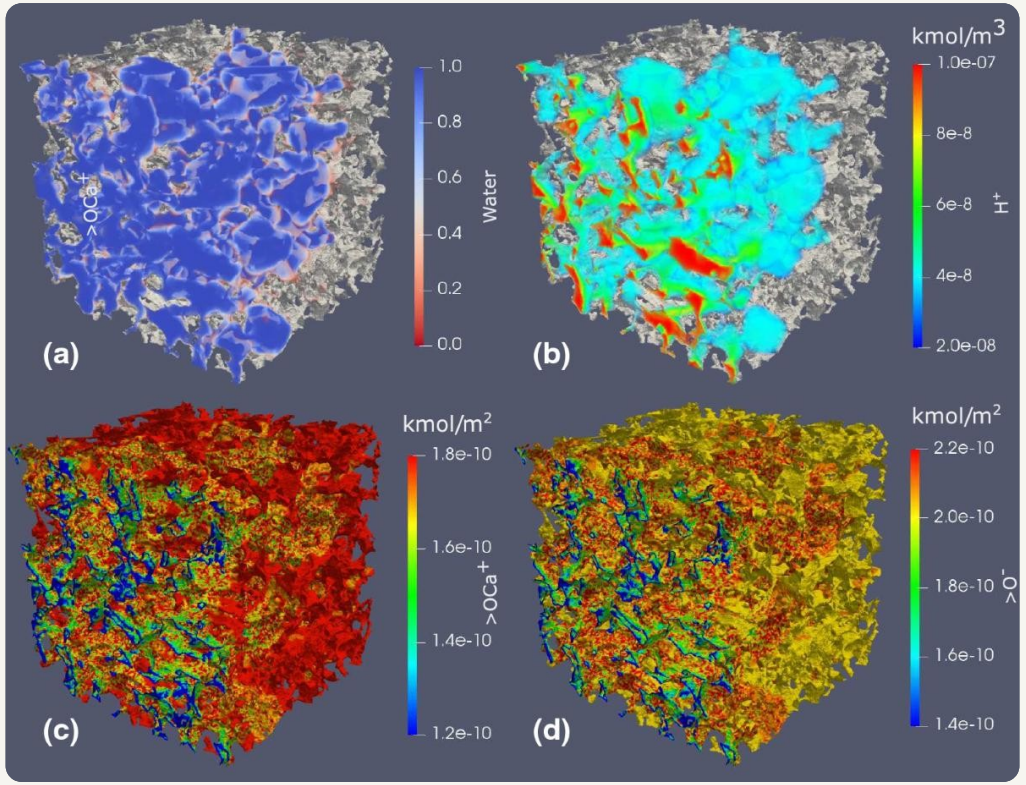
Mode	What the cation does	What would show it
Charge screening	Reduces repulsion between charged groups; links nothing	Surface potential shifts with ionic strength alone
Chelation	One ligand grips one ion at several points; still one chain	Coordination geometry, not network change
Bridging	One ion joins two separate chains, or a chain to a clay face	Rheology or retention changes beyond the ionic-strength effect

Denticity is not the same as connectivity: a monodentate contact can still be part of a bridge if the ion coordinates something else elsewhere. The clean contrast is a monovalent control against Ca^{2+} at matched ionic strength and pH.

Terminology after IUPAC Gold Book C01012 (chelation). A drop in surface potential alone would not establish cross-linking.

MODELLING METHOD · PUBLISHED SIMULATION

Pore-scale surface complexation could link clay chemistry to biofilm models



- Edge sites >SOH / >SO⁻ / >SOCa⁺; basal charge by cation exchange
- GeoChemFoam: OpenFOAM + PHREEQC on a micro-CT pore image
- Bentheimer sandstone with a montmorillonite surface model, $CaCl_2$ injected
- Next step: add biomass as a sorbent that changes over time

Sandstone, not bentonite. A method illustration only.

Figure: Maes & Menke 2021, Transp. Porous Media 139:271–299, Fig. 14, CC BY 4.0. Panels: (a) water, (b) H^+, (c) $>OCa^+$, (d) $>O^-$.

NATURAL ANALOGUE · QUALITATIVE

Oklo’s clay halo shows a sealed, reducing system works, but it is not bentonite

What it shows

- Uraninite, phosphates and bitumen held actinides and rare earths within metres for ~2 billion years
- The reactors made their own clay halo; ≈80% of the silica left reactor 2

What it doesn’t

- Halo is illite and Mg-chlorite formed at ~300–500 °C, not swelling smectite
- Cs, Sr, Ba almost all lost; no sorption or diffusion numbers for design

Gauthier-Lafaye, Holliger & Blanc 1996, GCA; Nagy et al. 1991, Nature; Oversby 2000, MRS Proc. (“qualitative, not performance-assessment numbers”).

MY OWN WORK · MODELS, NOT DATA

My Oklo models say heat binds before radiolysis, and they are models

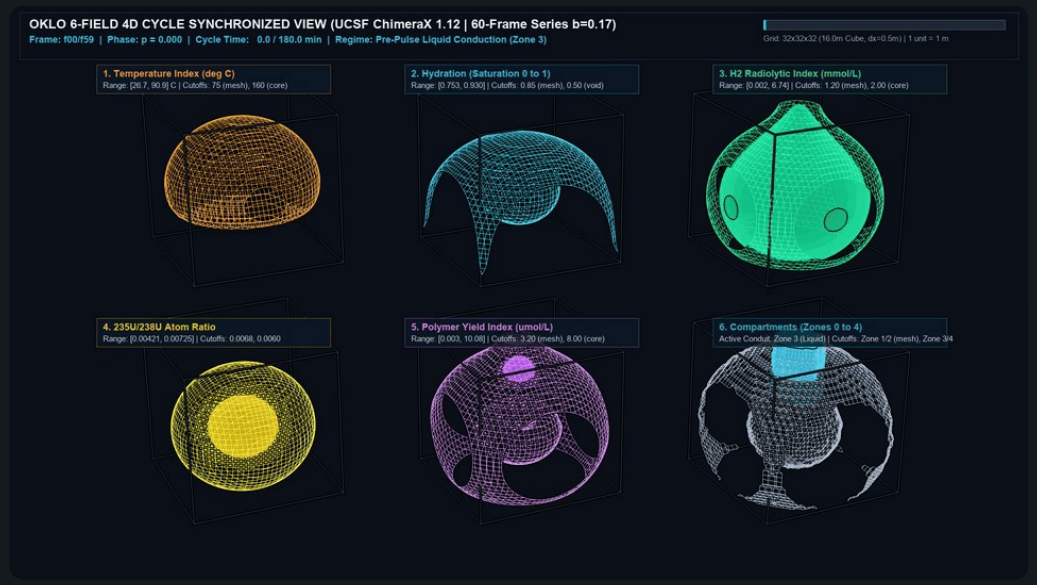
Type	Result
Model	Gel survives up to ≈0.05–12,000 Gy/h depending on repair rate (central ≈700 Gy/h)
Model	Gamma ≈2.5–6 Gy/h at the core face, mGy/h by 0.5 m; 122 °C limit lies 1.5–4 m out
Model	Catalase can raise the gamma ceiling by at most ≈26%; matrix turnover dominates
Measured	Heat gradient ≈100 °C/m near the cores; no biofilm evidence at the reaction zones

The bench pilot is what would turn the biggest assumptions (crosslink density, turnover, iron chemistry) into measurements.

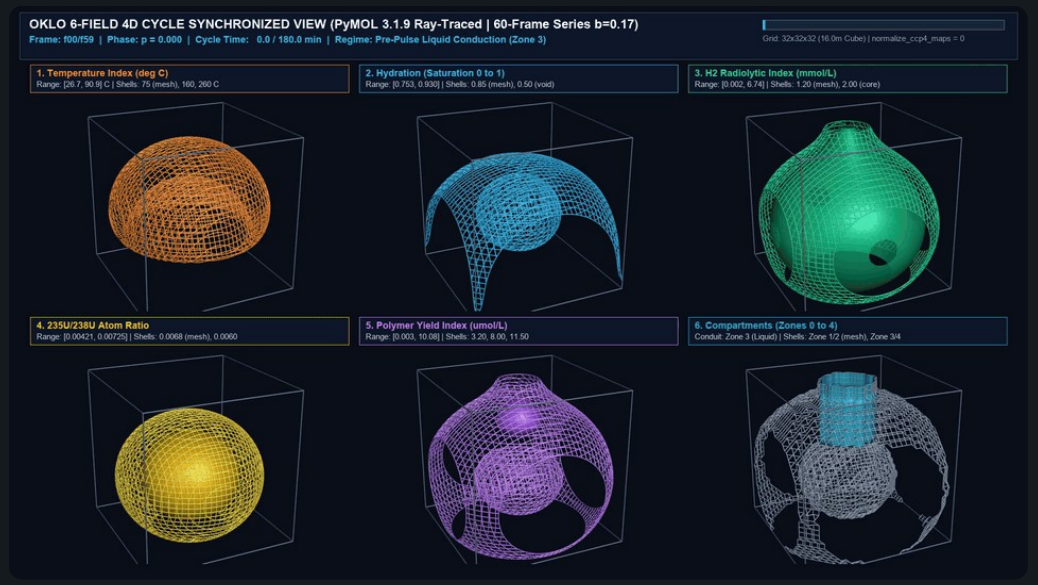
Measured values: Gauthier-Lafaye et al. 1996. Everything labelled “Model” is my own estimate from published parameters.

MODEL ILLUSTRATION · PRESCRIBED FIELDS, NOT DATA

What the six-field cycle looks like when you render it



UCSF ChimeraX · wireframe over core isosurface



PyMOL · ray-traced nested shells

What this is: six closed-form fields on a 32^3 grid, stepped through one cycle and rendered. The hollow shells are isosurfaces of those formulas. Nothing here is measured, fitted or simulated from transport equations.

Rendered from the frozen Revision-0 conformance fixture, which its own specification labels synthetic and prescribed-analytic. Shown as a visual aid only.

PROPOSED · NOT YET RUN

A 2–4 week pilot can show biotic versus abiotic, but not thresholds

Arm	Condition	What it answers
Slurry	MX-80 ≈1:10, live SRB, lactate and sulfate	Does sulfide form at all in this system?
Compacted	≈1.25 g/cm³ dry, live SRB	Does compaction change it at a reachable density?
Sterile clay	Autoclaved, same chemistry	Separates abiotic corrosion from microbial
No organism	Live clay, no inoculum	Tests the clay's own resident microbes

Readouts: weekly sulfide and pH; at the end SEM/EDS, Raman (mackinawite versus Cu_2S), XRD d001, and mass loss after standard cleaning.

Honest limit: at roughly 1 µm/yr, a few weeks of corrosion is below what a balance resolves. Sulfide and surface chemistry carry the result.

Design after SKB TR-16-09 and Martinez-Moreno et al. 2023, minus the ^{35}S radiotracer. Triplicate arms. All BSL-1, non-radioactive.

PROPOSED · ONLY IF THE PILOT WARRANTS IT

A 3–6 month study would bracket the density threshold

Core design

- Constant-volume cells at ≈1.75, 1.85, 1.95 g/cm³ wet, bracketing the published cut-off
- Sacrifice at 1, 3 and 6 months
- Add electrochemistry, sulfide profiles through the clay, and XPS

Optional arms

- An iron-reducing organism, to test whether clay iron chemistry shifts
- Magnetic susceptibility of the clay, a fast non-destructive check for biogenic magnetite
- Stable Co, Zn and Mn at ≈1–50 µM, split between pore water, clay, biomass and coupon by ICP
- No published precedent found for that partition design, so it is my inference

Densities after SKB TR-16-09. Magnetite versus siderite depends on headspace gas, pH and electron donor: Roh et al. 2003, Clays Clay Miner. 51:83-95. Stable metals only; no radioactive tracers at any stage.

SCOPE AND SEPARATION

The work stays BSL-1, non-radioactive and fully separate from my employer

Biosafety

- DSMZ lists the candidate sulfate reducer (DSM 644, = ATCC 29579) as Risk Group 1
- One supplier page carries a conflicting product label, so I will confirm with them and your IBC before ordering
- Note the genus was revised to *Nitratidesulfovibrio* in 2020
- IBC approval before any organism is handled

Separation

- No radioactivity anywhere in the design
- No employer facilities, data, imagery or branding
- Outside-activity disclosure in progress with my supervisor
- Everything here is public literature or my own labelled models

THE ASK

I bring modelling and analysis, and I would need bench access and mentorship

What I bring

- Reactive-transport and surface-complexation modelling
- Julia numerics and data analysis
- Analytical chemistry background: HPLC, ICP, GMP and GLP lab work
- A pre-registered readout plan, with every claim typed as measured, modelled or assumed

What I would need

- Anaerobic chamber and BSL-1 bench time
- Access to SEM/EDS, XRD and Raman
- Mentorship on biofilm and corrosion method choices
- Honest feedback on whether this is worth pursuing at all

Self-funded, evenings and weekends. Consumables for the pilot are plausibly a few thousand dollars, which is my estimate rather than a quote.

REVIEW RUBRIC · EXTERNAL SCORER

Four defect axes for reviewing a slide or protocol, without a Boltzmann factor

Axis	What it checks	What fails the check
1. Source	Primary-source attribution, units, and measured vs modelled labels	Secondary citations quoted as primary, missing density scales, or unverified DOIs
2. Coordination	Screening, chelation (IUPAC C01012), and bridging kept distinct	Inferring cross-linking from surface potential or denticity without a monovalent control
3. Boundary	Domain limits between analogue, model, and bentonite conditions	Treating Oklo chlorite/illite or prescribed 32^3 fields as bentonite transport data
4. Control	Sterile, killed-cell, and clay-free controls plus assay resolution	Missing abiotic baseline, or claiming mass-loss resolution over 2 to 4 weeks at ~1 µm/yr

Why drop the exponential: a self-graded revision that lowers its own defect score lands at P = 1.000 by arithmetic; the acceptance test never runs.

What makes the rubric work: an independent scorer who did not write the revision grades the four axes directly (checker is never the committer).

Four-axis review rubric (source, coordination, boundary, control) standing alone without statistical-mechanics weighting.

SOURCES

References, all public literature

- Ferrage et al. 2005a, Am. Mineral. 90:1358; 2005b, Clays Clay Miner. 53:348 — hydration states
- Greenspan 1977, J. Res. NBS — fixed humidity points
- Bengtsson et al., SKB TR-16-09 — density and sulfide
- Engel et al. 2023, mSphere — Grimsel MaCoTe, 5 years
- Bartak et al. 2024, World J Microbiol Biotechnol — review
- Necib et al. 2017, Swiss J Geosci — Mont Terri steel
- Burzan et al. 2022, Front Microbiol — Mont Terri microbes
- Ma et al. 2022, Appl Geochem — copper sulfide model
- Wersin et al. 2015 — Äspö ABM1 iron and clay
- Hadi et al. 2019 — FEBEX, 18 years
- Kim et al. 2004, Science — microbial smectite to illite
- Stucki & Kostka 2006 — iron reduction and clay properties
- Alimova et al. 2009 — biofilm polymers in clay interlayers
- Cuadros 2017, Clay Miner. — clay and microbe review
- Maes & Menke 2021, Transp. Porous Media — GeoChemFoam
- Gauthier-Lafaye et al. 1996, GCA; Oversby 2000 — Oklo

One citation is still unresolved and is flagged rather than quoted: a density threshold I can only trace to a secondary citation. Two earlier ambiguities are now closed: there are two separate Ferrage 2005 papers, and Masurat et al. 2010 is Appl. Clay Sci. 47:51-57.

APPENDIX A

Density and water-activity thresholds, with their basis

Value	Basis	Caution
Activity cut-off 1740–1880 kg/m³ wet	SKB TR-16-09, measured	Wet density; ≈1.2–1.4 dry
SKB design target 1950–2050 kg/m³ wet	Design guidance	Not a biological threshold
1.6 g/cm³ dry · a_w 0.96 · 2 MPa	Widely quoted triad	Only traced to a secondary citation; primary text unverified
a_w 0.954 at 1.50 g/cm³ dry	Grimsel field, measured	0.983 at 1.25 dry
Wet 2.0 ≈ dry 1.56 g/cm³	My conversion	Assumes grain density 2.75–2.78

APPENDIX B

Fixed humidity points for a hydration-controlled arm

Saturated salt at 25 °C	Relative humidity
Lithium chloride	11.3%
Potassium acetate	22.5%
Magnesium chloride	32.8%
Potassium carbonate	43.2%
Magnesium nitrate	52.9%
Sodium chloride	75.3%
Potassium sulfate	≈97.3%

Why it matters

Holding the clay at a chosen humidity sets how many water layers sit between the sheets, which XRD then reads directly as the d001 spacing.

That turns “hydration state” from an adjective into a controlled variable, which is the part I find most interesting.

Greenspan 1977, J. Res. National Bureau of Standards 81A:89.

PROPOSED MEASUREMENT LAYER · FLOW CYTOMETRY

Flow cytometry adds cell-state observations to the interface model

It measures suspended-cell phenotypes; by itself, it does not preserve where cells sat inside clay or a biofilm.

WHAT IT CAN MEASURE

- Gated cell-event counts and concentration using counting beads
- Dye-defined membrane-integrity classes, after strain-specific validation
- Scatter and per-cell fluorescence distributions
- Optional metabolic / ROS panels only after assay validation

WHAT IT CONSTRAINS

- Suspended and detached-cell abundance
- Fractions in measured physiological-state gates
- Net growth/death or attachment/detachment rates when time series are available
- Equifinality: similar bulk chemistry can arise from different cell-state histories

LIMITS

- Spatial position in the clay pore network
- EPS cross-link density or mineral phase
- Corrosion rate / causal mechanism
- “Live” status from fluorescence alone without controls; clay particles and aggregates can distort gates

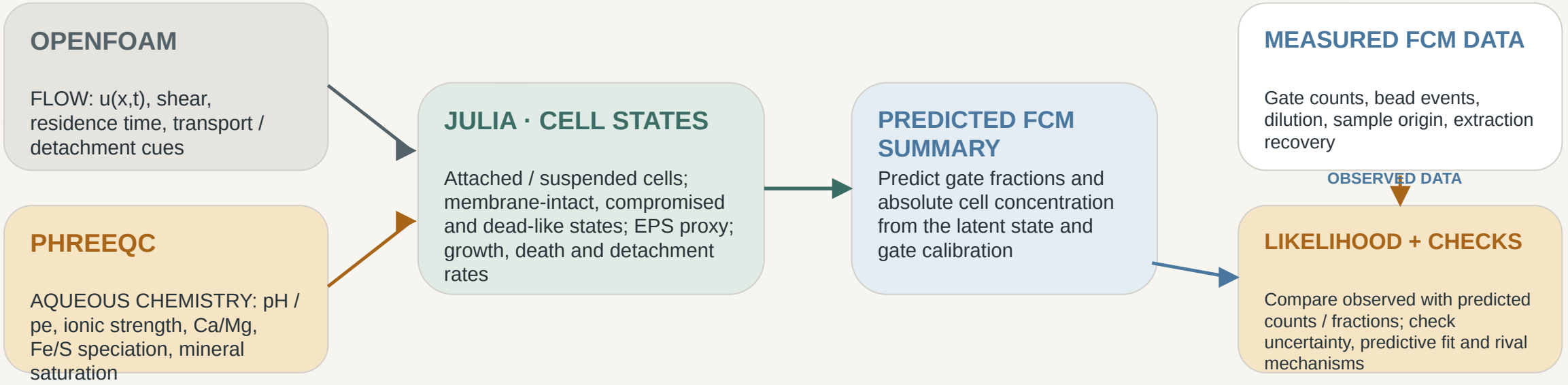
MINIMUM VIABLE ADD-ON: absolute counts + a validated membrane-integrity panel at preselected timepoints; keep maximum pit depth as the primary corrosion endpoint.

Sources: Kerstens et al. (2015), doi:10.1007/s12223-015-0400-4; Grainha et al. (2020), doi:10.3390/antibiotics9110741; Yip et al. (2022), doi:10.1371/journal.pcbi.1010533.

PROPOSED MODEL INTERFACE · NOT YET IMPLEMENTED

Keep the physics solvers; add a cytometry observation operator

OpenFOAM and PHREEQC drive the forward model; flow cytometry constrains the biological state through measured summaries.



BOUNDARY: cytometry loses spatial location; pair it with microscopy / endpoint surface analysis. Do not force OpenFOAM or PHREEQC into a nanocarrier assay unless those physical and aqueous-chemistry fields are genuinely part of that experiment.

Calibration framework: Yip et al. (2022), PLoS Comput Biol 18(10):e1010533. Lab context: Nano-Bio Lab, Single-Cell Analysis and computational cell classification.